MAPS_CANDIDATES

PREREG: MAPS_CANDIDATES — Residual ↔ Resonant candidate maps for Bridge C

Written 2026-10-24 ~5:05 PM CDT, before any test in this folder. Owner: Arvin Hampton. Parent folders (order2–order12, REVIEW_PARALLEL) read-only; new files only; CORE_FREEZE. All maps are **Category B** until proved. A pass is a *survived prediction*, never a proved Residual–Resonant bridge. Numerical near-miss ≠ pass. Runs: nice, OMP_NUM_THREADS=1, ulimit -v 3000000.

Sources for the residual side (verbatim facts only): oct17-master/README.md (T3 map; $|O_{res}| = B' = 539$), grok-chats/02-pure-math-status.md ($N_{tow}=243$, $N_{flux}=4880$, $f_{max}=21$, $Q=9$, $B'=\text{floor}((4880-21)/9)=539$; 56 same-tower faces), grok-chats/13-t3-vs-hqcc.md (K^+: $V=539$, $E=594$, $F=56$, $\beta=(1,0,0)$ over F2; fibre blocks 68/67; $L=18+521$), task brief (fibre partition $3\times68+5\times67$). Modular side: BRIDGE_C_35 REPORT/order11 (dim V=36, 105 components, $M^2=35P$, Lemma O, three unary thetas, $I=\tfrac{1}{2}\log(6+\sqrt{35})$).

Map-C1 (fundamental unit → packaging length)

Source: $\varepsilon = 6+\sqrt{35}$ (orthant angle $I = \tfrac{1}{2}\log\varepsilon$). Target: $L = 18+521 = 539$ packaging. Prediction (NEW): the multiplicative order of ε in $(\mathbb{Z}[\sqrt{35}]/539\mathbb{Z}[\sqrt{35}])^\times$ lies in {18, 521, 539}. PASS iff the exact order (integer arithmetic) is one of these three; else FAIL.

Map-C2 (three unary thetas → three T3 branches)

Source: the three weight-3/2 unary thetas with exponents $5n^2/2$, $R^2/34$, $35m^2/4$ ($n,R,m \in \mathbb{Z}$, as written). Target: T3 branches n mod 3 ∈ {0,1,2}. Prediction (NEW): with exponents scaled to integers by 68 ($170n^2$, $2R^2$, $595m^2$), each theta's exponents mod 3 put ≥ 2/3 of mass ($|\text{index}| \le 10^4$) on a single residue, and the three dominant residues are pairwise distinct (a bijection theta ↔ branch). PASS iff both hold; else FAIL.

Map-C3 (P-parity on the 105 companions → K^+ F2 cell counts)

Source: P ($M^2=35P$) acting on V (dim 36). Target: K^+ counts {V=539, E=594, F=56} / β over F2. Prediction: dim V^{P=−1} − dim V^{P=+1} ≡ $\chi(K^+)=1$ (mod 2), and dim V^{P=−1} ∈ {1, 18, 21, 56 mod 36 = 20}. PASS iff both. Test requires the exact P on V (SF/A35_lawspace_exact). If not cheaply available → OPEN.

Map-C4 (Kronecker character of Q(√35) → eight fibre blocks)

Source: χ_D, D=140 (discriminant of Q(√35)); the Lemma-O sector sign is a quadratic character. Target: O_res indexed 1..539 in order, cut consecutively into blocks of sizes 68,68,68,67,67,67,67,67 (interpretation fixed now). Prediction (NEW): every block sum $S_b = \Sigma_{n\in b} \chi_{140}(n)$ satisfies $|S_b| \le 1$, and the 68-blocks and 67-blocks have opposite sums' sign pattern (all 68-block sums ≥0, all 67-block sums ≤0 or vice versa). PASS iff both; else FAIL.

Map-C5 (wall Q → T3 free orbit of N_flux)

Source: walls Q ∈ {1, 7/2, 595/2}. Target: free T3 orbit of 4880 (14 steps, known). Prediction (NEW): the orbit of 4880 contains an element n with 2n ∈ {2, 7, 595} other than the attractor 1, i.e. it passes through 595 (=2·595/2) or hits a value ≡ 595 mod 539·? — fixed sharp form: **595 ∈ orbit(4880)**. PASS iff 595 occurs in the free T3 orbit of 4880; else FAIL.

Map-C6 (resonant path Π → modular S-law)

Source: S-law matrix M_S (order 4). Target: resonant path Π, length 539. Prediction: the 539-step resonant word, read mod 4, has M_S-order profile 539 ≡ 3 (mod 4) ⇒ the path's end-state is S^3-related to start. Requires a written Π trajectory (not in repo) → OPEN unless found. # RESULTS: MAPS_CANDIDATES PREREG.md sha256 fdd4c2be09aef174ab85bb5d7b72cce3cedc9d498b7f47d72814e90092dfa424 (hashed before tests). Script: scripts/tests.py (hash in scripts/SHA256SUMS). Data: data/results.json. Exact integer arithmetic; runtime < 1 s; well under 3 GB.

Map	Prediction	Measured	Verdict	Category
C1	ord(6+√35) mod 539 ∈ {18,521,539}	$294 = 2\cdot3\cdot7^2$	**FAIL**	B (candidate dead)
C2	3 thetas → 3 distinct dominant residues mod 3	$170n^2,2R^2$ → dominant 2 (2/3); $595m^2$ → dominant 1; residues {2,2,1} not distinct	**FAIL**	B (dead)

Map	Prediction	Measured	Verdict	Category
C3	P-eigen dims on V vs K^+ counts	not measured	**OPEN**	B
C4	χ_{140} block sums	S_b	≤ 1, opposite signs 68 vs 67 blocks	[1,−1,0,0,0,−1,3,−5]
C5	595 ∈ free T3 orbit of 4880	orbit $4880 \to \ldots \to 1$ (14 steps) avoids 595	**FAIL**	B (dead)
C6	Π end-state S^3-related (539 ≡ 3 mod 4)	no written Π trajectory found	**OPEN**	B

No Residual–Resonant bridge is claimed. The order 294 in C1 is a post-hoc observation only (not predicted; it is not 18, 521 or 539) and is not upgraded. # NOTES - C2 FAIL is structural: 170 ≡ 2 ≡ 2 (mod 3), so the $5n^2/2$ and $R^2/34$ thetas share a residue profile under the ×68 scaling; any theta ↔ branch bijection by this route is impossible. - C4 uses a fixed interpretation (O_res indexed 1..539, consecutive blocks 68,68,68,67×5). A different written ordering of O_res would be a NEW prereg, not a rescue of C4. - C5 note: T3 orbit of 4880 is the known 14-step free orbit; C5 tested only whether it meets 595. - C6 is weakly motivated (mod-4 numerology on $M_S^4 = I$); keep as lowest priority. - Dropped as too vague to make falsifiable: $\log(6+\sqrt{35})$ vs $N_flux = \text{floor}(e^3 \cdot 243)$ (no mechanism); 67/68 as targets (Arvin: consequences of eight blocks). - C3 next measurement: from `_git_v8_unpushed_clone/reports/s_family_paths/A35_lawspace_exact` (exact_main.json, X_modp_*.npy), build $P = M_S^2/35$ on the 36-dim V mod p and compute rank($P \mp I$) at ≥2 primes. Pass/fail as in PREREG. - C6 next measurement: a written, hashed Π (539-step resonant word) from residual sources; none located in repo.